Adilet Shynggys

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RESEARCH INTERESTS

Heat transfer, thermodynamics, and two-phase flow for cleaner, more efficient, and safer power generation, particularly nuclear energy: how microscale surface geometry, texture, and wettability govern the supply of liquid to heated surfaces, rewetting, boiling, and the boiling crisis, and how those effects carry to reactor thermal-hydraulics.

EDUCATION

Expected June 2027

B.Eng., Mechanical and Aerospace Engineering

Nazarbayev University • Astana, Kazakhstan

- GPA **3.86 / 4.00**; **ranked 1st** in cohort; Dean's List, four consecutive semesters.
- Coursework: Engineering Thermodynamics, Fluid Mechanics, Heat and Mass Transfer (in progress), Structural Mechanics, Materials and Manufacturing, Numerical Methods, Engineering Dynamics I–II.

Spring 2026

Visiting Exchange Student, Mechanical Engineering

The University of Hong Kong • Hong Kong SAR

- Energy Conversion Systems, Design and Manufacture, Additive Manufacturing; collaborated with Prof. Wei Yan on curriculum design for an undergraduate course.

RESEARCH AND DESIGN

Aug 2026 – Present

Capillary-Driven Surface Engineering

Dept. of Mechanical and Aerospace Engineering, Nazarbayev University
Advisor: Prof. Amgad Salama

- Studying how microchannel taper geometry moves liquid with no pump: the taper creates a Laplace-pressure asymmetry that should drive wetting liquids toward the narrow end and non-wetting liquids toward the wide end.
- With the group, designed the inverted photomask and completed the full clean-room photolithography process, producing a first successful sample of tapered SU-8 microchannels on silicon.
- Next: transport experiments across taper geometries and droplet- and slug-velocity measurement, toward understanding how geometry speeds the rewetting of heated surfaces.

Aug 2026 – Present

Senior Capstone (Engineering Design): VVER-1200 Reactor Core

Nazarbayev University • Team of four
Advisor: Prof. Amgad Salama

- Converted the required electrical output to 3,200 MWth of core thermal power via cycle efficiency, and used a target burnup near 50 GWd/tHM to size fuel pellets, rods, and bundles into a hexagonal core configuration.
- Now entering thermal-hydraulic analysis: coolant flow and temperature rise, critical heat flux, and margins to subcooled boiling and DNB, feeding a reduced-order model to be validated against published reference data.

Jan – Dec 2025

Triboelectric Nanogenerators for Wearable Sensing

Energy Conversion Laboratory, Nazarbayev University
Supervisor: Dr. Shabnam Yavari

- Fabricated triboelectric nanogenerators with the group by electrospinning and silicone casting, raising output power density by 15–20%.
- Ran device characterization across humidity and cyclic durability, and built the electrical workflow behind it: load sweeps, data acquisition, and analysis in OriginPro and MATLAB.
- Results contributed to a co-authored article in *Small* and a first-author poster awarded **Best Research Poster**.

PUBLICATIONS AND PRESENTATIONS

[1] S. Faramarzi, S. Yavari, U. Ospanova, **A. Shynggys**, et al., "Polarity-Engineered, Tribo-Positive Imidazolium-Mediated Crosslinked 6FDA-Durene Films for Triboelectric Nanogenerator," *Small*, 2026, e74486. doi:10.1002/smll.74486

[2] **A. Shynggys**, S. Yavari, G. Kalimuldina, "Green-Synthesized Ag Nanoparticles Incorporated into High-Performance Triboelectric Wearable Sensors." Poster, Belt and Road Initiative Conference, Astana, 2025 (**Best Research Poster**); abstract, INESS-2025, Astana, Aug. 2025.

TEACHING, LEADERSHIP, SERVICE

Elected Undergraduate Representative

Student Advisory Body, School of Engineering, Nazarbayev University

2026 – Present

- One of two students elected for the major, one undergraduate and one graduate, carrying student concerns and recommendations directly to the Dean and Vice Dean.

Cohort Representative, Mechanical and Aerospace Engineering

2024 – 2026

Nazarbayev University

- Ran **10+** exam-preparation nights for the cohort.
- Raised student requests with the dean, leading to a rescheduled exam and more exchange opportunities.

Physics Tutor (volunteer)

2023 – 2024

*Nazarbayev University • tutored **20+** first-year students*

Community Service

2025 – 2026

- Sponsorship team, charity trail run supporting MamaPro, a nonprofit providing business education to single mothers: **150** runners, **600,000 KZT** (∼$1,300) raised.
- Volunteer, Mereke Charity Foundation (retirement-home visits, winter clothing drive for children from vulnerable families) and HKU Muslim Society (organizing iftars during Ramadan).

OTHER DESIGN PROJECTS

- *Multi-stage gearbox* (University of Hong Kong): sized shafts by the ASME bending–torsion criterion and selected gears, keyways, and bearings for the required fatigue life.
- *Interactive art installation* (HKU, with Prof. Homayoon Kazerooni, UC Berkeley): modeled petal forms in SolidWorks, checked collisions and stability, and 3D-printed each piece.

INDUSTRY EXPERIENCE

Product Support Engineering Intern

Wabtec Corporation, Kazakhstan • awarded as the Mechaton 1st-place prize

June – July 2025

- Analyzed traction-motor thermal performance and supported root-cause review of temperature excursions.
- Audited shop-floor manufacturing against drawings, reconciling GOST with ASME, and traced an insulation inconsistency to different export-market climates.

AWARDS AND HONORS

Presidential Scholarship of the Republic of Kazakhstan (2026–27) | Yessenov Foundation Scholarship, 20 of 500+ applicants (2026–27) | NU President's Scholarship (2024–25) | Best Research Poster, Belt and Road Initiative Conference (2025) | 1st Place, Mechaton Engineering Case Competition (2025)

SKILLS

Computing: MATLAB, Python, Mathematica, OriginPro, LaTeX | **CAD:** SolidWorks (CSWA certified) | **Lab:** photolithography, electrospinning, silicone casting, 3D printing, device characterization | **Languages:** Kazakh (native), Russian (fluent), English (IELTS 8.0)